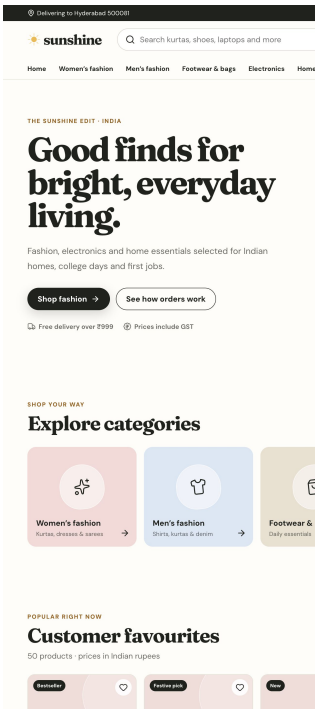
PORTFOLIO PROJECT REPORT

Sunshine

Agent-based Indian retail across frontend, Java backend and Python services



Author	Divya Rachala
Portfolio focus	Frontend + Java backend + Python backend/data
Project type	Final-year student-scale retail system
Region	India-first catalogue, pricing and delivery flow
Date	August 2026

A complete shopper journey from product discovery to a traceable order result. All products, payments and order data are synthetic demonstrations.

01 / EXECUTIVE SUMMARY

Project purpose and scope

Sunshine was designed to demonstrate a connected system rather than a collection of unrelated technologies.

Problem statement

A basic product grid does not show what happens after a shopper clicks buy. Sunshine models the complete decision path: stock reservation, payment outcome, delivery planning, customer feedback and a visible operational trace.

Portfolio objectives

- Build a clean, responsive Indian retail experience with prices in INR and familiar delivery expectations.
- Use Java for typed order orchestration and failure handling.
- Use Python for a FastAPI recommendation endpoint and a pandas retail KPI workflow.
- Demonstrate five bounded agents without misrepresenting them as generative AI.
- Deploy a working public demo while documenting the boundary between Vercel and local services.

Delivered scope

Area	Delivered capability	Evidence
Catalogue	50 synthetic products in five retail categories	Static data + 50 generated product pages
Customer flow	Search, product, cart, checkout, profile and orders	Browser-tested repeat-visit paths
Java	Five bounded order agents	Spring Boot + three JUnit scenarios
Python	Recommendation API and retail KPI script	FastAPI + pandas + four pytest checks
Delivery	Docker Compose, CI and Vercel-ready UI	Three GitHub Actions jobs

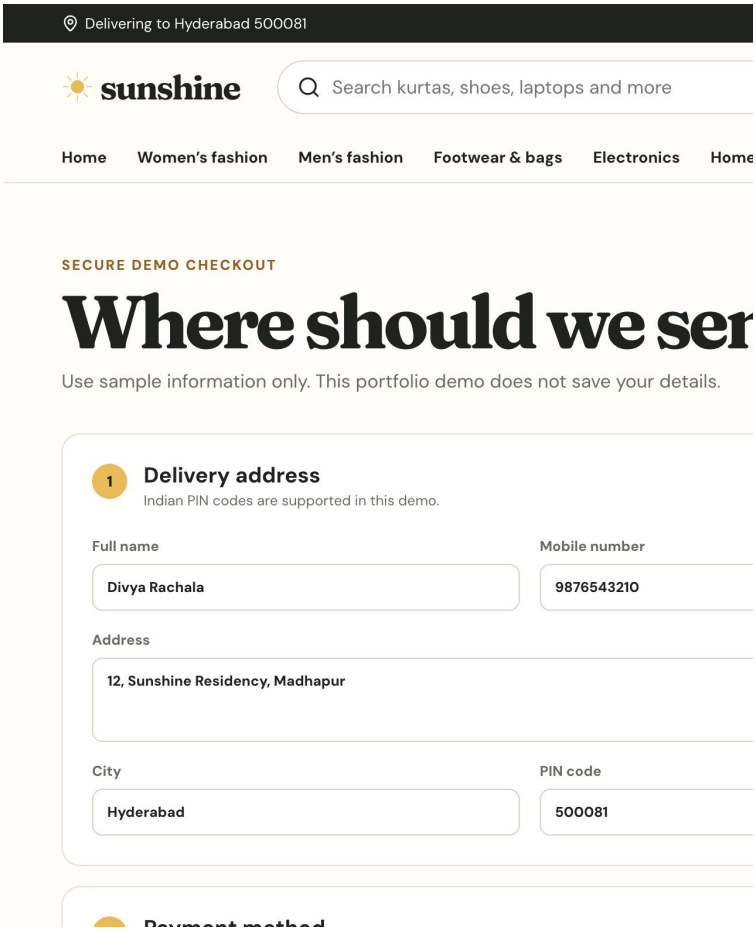
The scope deliberately excludes authentication, a production database and a real payment gateway. Those choices keep the project explainable and credible for a final-year student.

02 / CUSTOMER JOURNEY

From discovery to delivery

The interface follows a realistic retail sequence while keeping every payment and order operation safely simulated.

Step	Customer action	System response
1	Search or choose a category	Filters the 50-product catalogue
2	Open a product and choose size	Shows pricing, delivery and recommendations
3	Add item and adjust quantity	Persists cart in versioned local storage
4	Enter address and payment method	Validates Indian mobile and PIN formats
5	Choose a demonstration scenario	Runs success, decline or stock failure
6	Review order and profile	Persists history, inventory and agent trace



Retail rules

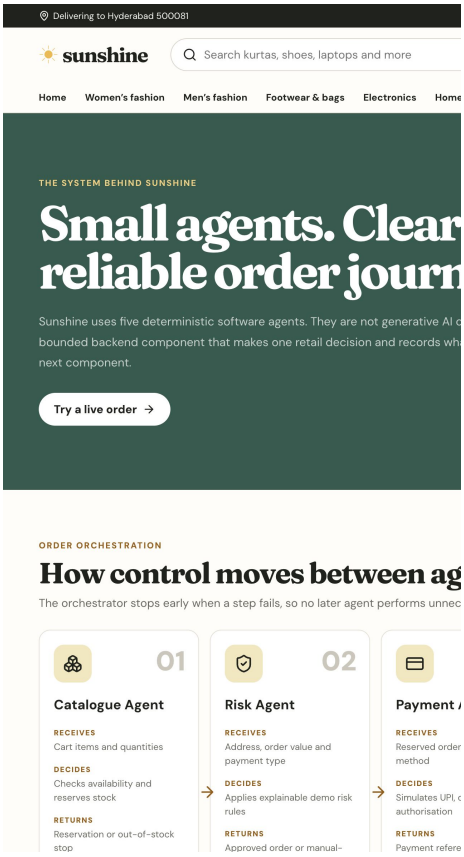
- Free delivery starts at INR 999; smaller carts add INR 79.
- UPI, card and cash on delivery are available as simulated methods.
- A successful order reduces browser stock; a decline keeps stock available for retry.

03 / SYSTEM ARCHITECTURE

One browser contract, three technology stacks

The Next.js route handlers provide stable browser endpoints and proxy to Java or Python when their service URLs are configured.

Layer	Technology	Responsibility
Browser	React + TypeScript	Interactive catalogue, cart and checkout state
Web app	Next.js 16	Pages, metadata, route handlers and Vercel deployment
Order service	Java 17 + Spring Boot 4	Validation and five-agent orchestration
Recommendation service	Python 3.13 + FastAPI	Content-based product ranking
Analytics	pandas	Raw order CSV to KPI JSON
Delivery	Docker + GitHub Actions	Reproducible local stack and automated checks



Deployment adapter pattern

If JAVA_BACKEND_URL or PYTHON_BACKEND_URL is present, Next.js proxies to the real service. If a service is unavailable, a TypeScript adapter applies the same deterministic contract so the Vercel demo remains usable. The response identifies the source, which allowed integration tests to prove that Java and Python were actually called locally.

Bounded agents and explicit failure paths

Spring Boot is the canonical order service. The orchestrator calls agents in sequence and stops safely when a prerequisite fails.

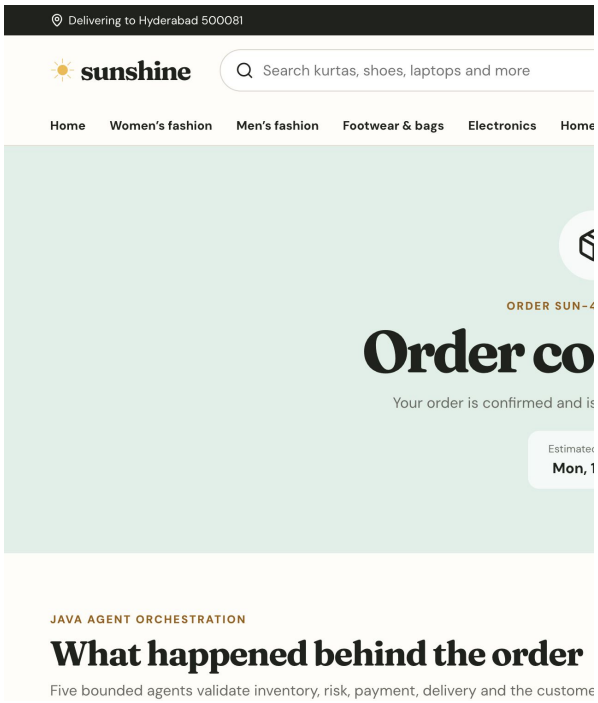
Agent	Input	Decision	Output
Catalogue	Cart + stock	Can stock be reserved?	Reservation or OUT_OF_STOCK
Risk	Order context	Do rules pass?	Approved decision
Payment	Reserved order	Can payment proceed?	Reference or failure
Fulfilment	Eligible order	When can it arrive?	Shipment plan
Notification	Final result	Which update applies?	History event

Why agents instead of one large service?

Each class owns one reason to change. Inventory rules can evolve without changing delivery logic; payment methods can be extended without rewriting the catalogue check. The orchestrator remains small because it coordinates results rather than implementing every business rule.

Scenario behaviour

Scenario	Catalogue	Risk	Payment	Delivery	Notify
SUCCESS	done	done	done	done	done
PAYMENT_FAILED	done	done	failed	skipped	done
OUT_OF_STOCK	failed	skipped	skipped	skipped	done



The Java API returned the custom response header x-sunshine-service: java during local integration testing.

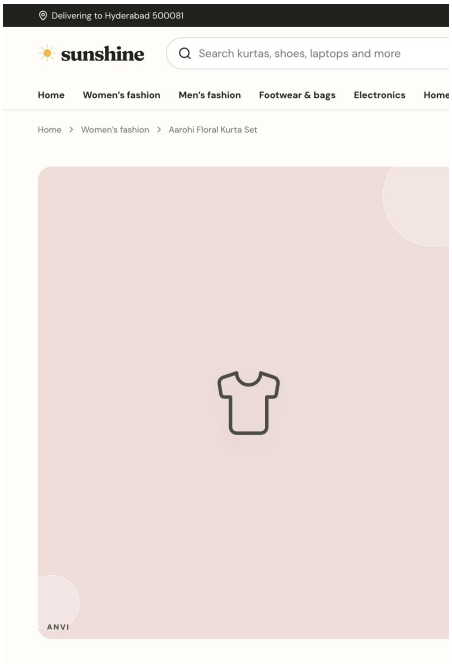
Recommendations and measurable retail KPIs

Python has two focused roles: ranking related products through FastAPI and transforming a raw order CSV through pandas.

Recommendation method

The service receives a reference product and same-contract candidates from Next.js. It filters to the same category and ranks by rating and relative price distance. This is intentionally content-based because the demo does not collect customer histories.

score = rating x 2 - absolute price difference / reference price



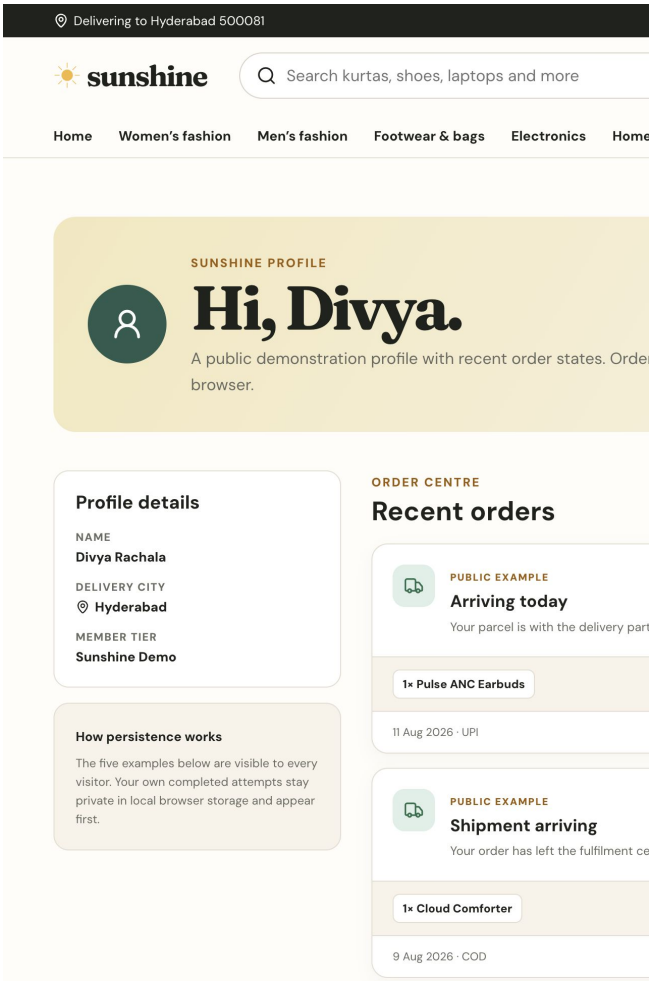
KPI pipeline output

KPI	Result
Orders received	12
Orders confirmed	9
Confirmation rate	75.0%
Confirmed revenue	INR 37,136
Average order value	INR 4,126.22
Top revenue city / category	Delhi / Electronics

06 / FRONTEND AND ORDER HISTORY

Retail clarity with repeat-visit state

Sunshine combines an original retail identity with a profile, seeded public order examples and private browser-persisted activity.



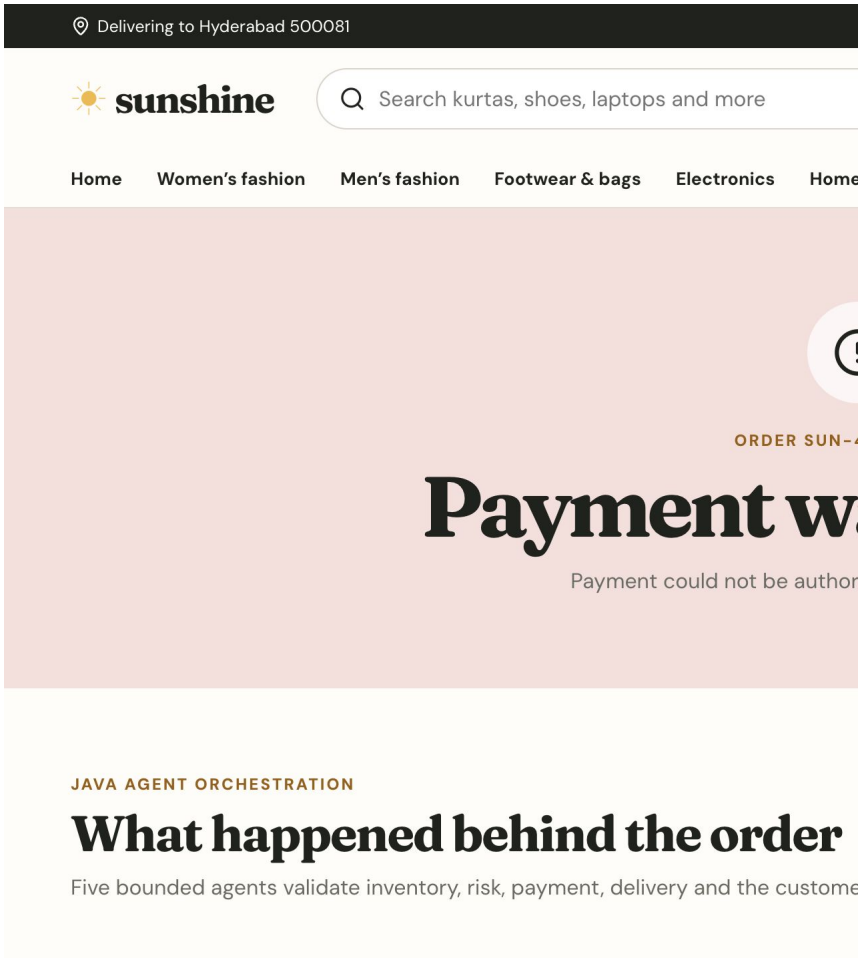
Decision	Reason
Warm cream, yellow and green palette	Creates a recognisable Sunshine identity without imitating Amazon
Seeded order lifecycle examples	Every visitor can inspect arriving, shipped, delivered and failed states
Versioned browser storage	Personal orders and stock changes persist privately across visits
Low-stock product rules	The last-unit purchase becomes unavailable when the shopper returns
Responsive grids	Profile and orders collapse cleanly on small screens

Accessibility checks include labelled search, semantic navigation, form labels, radio groups, keyboard-reachable buttons, visible focus styles and readable success/failure text that does not depend only on colour.

07 / FAILURE HANDLING

A failed payment is a designed outcome

A portfolio system should explain what happens when an operation does not succeed, not hide every failure behind a generic error message.



Observed behaviour

- The catalogue reservation completes before payment begins.
- The Payment Agent returns failed with a method-specific explanation.
- The Fulfilment Agent is marked skipped because payment eligibility was not met.
- No payment reference or delivery date is generated.
- The cart remains available so the customer can retry without rebuilding it.
- The Notification Agent records the failure in the visitor's recent-order history.

This trace turns a failure into evidence of control flow, state handling and customer communication.

08 / QUALITY AND DELIVERY

Automated checks across the complete stack

Tests target business behaviour rather than only verifying that files compile.

Quality gate	Coverage	Result
Vitest	Pricing, delivery fee and stock-aware adapter paths	7 passed
JUnit	Success, decline, scenario and real stock exhaustion	4 passed
pytest	Ranking category, determinism, API and validation	4 passed
ESLint	React and TypeScript quality rules	Passed
Next.js build	Routes, types, profile and 50 product paths	60 pages generated
Browser QA	Profile, last unit, return visit and failures	Passed
Integration	Next.js -> Java and Next.js -> Python	Both sources confirmed

Continuous integration

GitHub Actions runs three independent jobs. The frontend job installs Node 22 dependencies, tests, lints and builds. The Java job sets up Temurin 17 and runs Maven Wrapper tests. The Python job installs versioned requirements and runs pytest.

Deployment

- Docker Compose starts the Next.js, Java and Python services as one local system.
- Vercel hosts the public Next.js interface and compatible route adapters.
- Environment variables switch the route handlers to external Java and Python services without browser changes.

Environment variable	Purpose
JAVA_BACKEND_URL	Proxy order requests to Spring Boot
PYTHON_BACKEND_URL	Proxy recommendation scoring to FastAPI

Security boundary

The demo never requests real card, UPI, password or customer identity data. The checkout form explicitly asks for sample information and does not persist delivery details.

09 / REFLECTION

What this project proves and what comes next

The strongest part of Sunshine is not any single framework; it is the connection between customer state, service contracts, failure rules and verifiable outputs.

Skills demonstrated

Interview topic	Concrete example to discuss
React state	Versioned cart, order and inventory stores with useSyncExternalStore
Next.js	Server-rendered product paths plus small client boundaries
Java design	Five constructor-injected agents coordinated by OrderOrchestrator
Python API	Typed FastAPI request model and deterministic ranking
Data analysis	CSV-to-JSON KPI pipeline with pandas
Failure handling	Early exit, skipped steps and retry-safe cart state
Deployment	Contract-preserving adapter for platform runtime limits

Limitations

- Synthetic data cannot prove real customer demand or production-scale performance.
- The application has no authentication, shared order database or live stock source.
- Content-based ranking does not learn from user behaviour.
- Illustrative agent durations are trace labels, not benchmarks.

Reasonable next steps

- Add PostgreSQL order persistence and idempotency keys.
- Deploy Spring Boot and FastAPI to suitable container runtimes and configure Vercel URLs.
- Introduce a real event log and observable correlation ID across services.
- Replace synthetic recommendations only when consented behavioural data is available.

Sunshine is intentionally a finished, explainable student project rather than an unfinished enterprise platform.

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